

Apex Institute of Technology

Department of Mathematics — MATH 204

Multivariable Calculus & Linear Algebra

Worksheet 04

Topic: Spectral Theorem &
Vector Fields

Term: Fall 2026

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1. Linear Transformations & Eigenvalue Problems

Problem 1.1: Symmetric Matrix Diagonalization & Spectral Decomposition

Consider the symmetric matrix $A \in \mathbb{R}^{3 \times 3}$ defined by:

$$A = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{bmatrix}$$

- Find the characteristic polynomial $p_A(\lambda) = \det(A - \lambda I)$ and derive all eigenvalues.
- Construct an orthonormal basis of eigenvectors for $\mathbb{R}^3$ and matrix P such that $P^T A P = D$.
- Express A^5 using the spectral decomposition $A = \sum_i \lambda_i \mathbf{u}_i \mathbf{u}_i^T$.

Note: A real symmetric matrix $A = A^T$ has orthogonal eigenvectors for distinct eigenvalues. This guarantees diagonalizability over $\mathbb{R}$.

Detailed Derivation Step-by-Step:

Step 1: Compute eigenvalues. Subtract λ along the main diagonal:

$$\det(A - \lambda I) = \det \begin{bmatrix} 3-\lambda & 1 & 1 \\ 1 & 3-\lambda & 1 \\ 1 & 1 & 3-\lambda \end{bmatrix} = (2-\lambda) \det \begin{bmatrix} 1 & 0 & 0 \\ 1 & 4-\lambda & 1 \\ 1 & 2 & 3-\lambda \end{bmatrix}$$

Expanding the remaining 2×2 determinant:

$$(4-\lambda)(3-\lambda) - 2 = \lambda^2 - 7\lambda + 10 = (\lambda-2)(\lambda-5) \implies p_A(\lambda) = -(\lambda-2)^2(\lambda-5) = 0$$

Thus, the eigenvalues are $\lambda_1 = 5$ (mult. 1) and $\lambda_2 = \lambda_3 = 2$ (mult. 2).**Step 2: Eigenspace for $\lambda_1 = 5$.**

$$(A - 5I)\mathbf{v} = 0 \implies -2x + y + z = 0, \quad x - 2y + z = 0 \implies \mathbf{u}_1 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Step 3: Eigenspace for $\lambda_2 = 2$.

$$(A - 2I)\mathbf{v} = 0 \implies x + y + z = 0 \implies \mathbf{u}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \quad \mathbf{u}_3 = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$$

Algebraic Multiplicity

Multiplicity of λ as a root of $\det(A - \lambda I) = 0$. Geometric multiplicity satisfies $1 \leq \dim(\ker(A - \lambda I)) \leq \text{alg. mult.}$